

JEE MAIN REVISION MASTER: MATHEMATICAL REASONING

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Logical Statements, Connectives, Contrapositive & Tautology	9 PYQs	Mathematical Logic

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Logical Statements, Connectives, Contrapositive & Tautology	$\text{Contrapositive of } (p \implies q) \equiv (\sim q \implies \sim p), \quad \sim(p \implies q) \equiv p \wedge \sim q, \quad \sim(p \vee q) \equiv \sim p \wedge \sim q$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Logical Statements, Connectives, Contrapositive & Tautology (High Yield — 9 PYQs)

Core Principle: Logical statements and truth values: Negation ($\sim p$), Conjunction ($p \wedge q$, true only if both true), Disjunction ($p \vee q$, false only if both false). Conditional $p \implies q \equiv \sim p \vee q$. Its Contrapositive is $\sim q \implies \sim p$ (logically equivalent to $p \implies q$). Converse is $q \implies p$; Inverse is $\sim p \implies \sim q$. Negation of conditional: $\sim(p \implies q) \equiv p \wedge \sim q$. Tautology: always true; Contradiction: always false.

■ **Caution / JEE Pitfall:** The contrapositive of 'If p, then q' is 'If NOT q, then NOT p'. It is logically equivalent to the original statement, unlike converse or inverse!

■ **Speed Trick:** To find contrapositive in English: Swap the 'if' and 'then' clauses and negate both.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The solution of $3(x-1) \leq 2(x-3)$ is..." [Ans: A]
- [2024 Shift] Q#2 (MCQ): "The negation of the statement "For every real number x ; x^2+5 is positive" is..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "The contrapositive of the statement. "If two lines do not intersect in the same plane, then they are parallel." is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The solution of $3(x-1) \leq 2(x-3)$ is

- (A) $x \leq -3$ **[Correct]**
- (B) $x \geq -3$
- (C) $x \leq 3$
- (D) $x \geq 3$

Correct Answer: A

Solution Analysis: Given inequality: $3(x-1) \leq 2(x-3)$ Open the brackets: $3x-3 \leq 2x-6$ Now subtract $2x$ from both sides: $x-3 \leq -6$ Add 3 to both sides: $x \leq -3$ So, the solution is: $x \leq -3$ Hence, the correct option is Option A .